Unsupervised SAR Image Change Detection Based on SIFT Keypoints and Region Information

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***Abstract*—This letter presents a new unsupervised distribution-free change detection method for synthetic aperture radar (SAR) images based on scale-invariant feature transform (SIFT) keypoints and region information. Since the SIFT can detect blob-like structures in an image and be insensitive to noise, we first extract noise-robust SIFT keypoints in the log-ratio image to reduce the detection range. Then, in order to obtain accurate changed regions, rather than directly obtaining the change-detection map from the difference image as in some traditional change detection methods, we make segmentation around the extracted keypoints in the two original multitemporal SAR images, where the edges of detection regions are much clearer than those in the difference image, and further compare the two segmentations to generate the change-detection map. This method utilizes the bloblike structure information offered by SIFT keypoints and the region information extracted via image segmentation. Experiments on real SAR images demonstrate the effectiveness of the proposed method.**

***Index Terms*—Change detection, scale-invariant feature transform (SIFT), segmentation, synthetic aperture radar (SAR) images.**

I. Introduction

CHANGE detection on synthetic aperture radar (SAR) images is a process that utilizes a pair of images acquired over the same geographical area at different times to identify the changes that may have occurred between the considered acquisition dates. Due to the insensitiveness of SAR sensors to atmosphere (particularly rain and clouds), the SAR system is more suitable than other remote sensing devices employed in change detection. Hence, in the past decades, it has been successfully used in many applications such as agricultural surveys, environmental monitoring, damage assessment, urban studies, and forest monitoring. However, change detection on SAR images encounters more difficulties than others due to the effect of the speckle noise [1], [2].

SAR image change detection techniques can be categorized into two main research streams: supervised and unsupervised paradigms. In recent years, some semisupervised methods are also proposed [3]. However, unsupervised methods have been widely studied since they do not need any prior knowledge of images. This letter focuses the attention on unsupervised SAR image change detection.

With the development of the statistical theory, modeling and analyzing the observable data is becoming an effective way to achieve automatic change detection [4]–[7]. In [4], an unsupervised approach to automatic change detection is proposed based on the statistical modeling. It models the ratio image as a mixture of two generalized Gaussian distributions associated with changed and unchanged classes and employs an automatic threshold selection method to obtain the change map. However, owing to the effect of speckle noise in SAR images, the pixels in changed areas may share a wide range of values with the unchanged pixels. As a result, the selection of the best threshold values is a challenging task [2]. Moreover, the accuracy will decrease when the assumed statistical model does not match the practical situation.

To overcome the disadvantages of the statistical modeling-based methods, some distribution-free change detection methods are proposed. Most of these methods are based on clustering algorithms [8]–[10]. In [8], the image fusion technique is introduced to generate a different image by using complementary information from a mean-ratio image and a log-ratio image. In order to overcome the sensitiveness of the traditional fuzzy C-means (FCM) algorithm to speckle noise, a reformulated fuzzy local-information C-means (RFLICM) clustering algorithm is also proposed in [8] to classify the changed and unchanged regions in the fused difference image. Although the RFLICM clustering algorithm can suppress the noise to a certain extent, the cost of computation is considerably high.

Previous works on SAR image change detection mainly concentrated on using the pixel information. However, owing to the speckle noise in the SAR image, the methods based on pixel comparison may be often inefficient. In this letter, we propose a novel method which exploits not only the scale-invariant feature transform (SIFT) keypoints extracted from the difference image to improve the detection robustness to the speckle noise but also the region information around the SIFT keypoints in the original SAR images to obtain accurate changed regions. The detailed procedure is summarized as follows. The log-ratio image is first generated from the two multitemporal images. Then, since the SIFT feature can detect bloblike structures in an image and be insensitive to noise [11], the noise-robust SIFT keypoints are extracted in the log-ratio image to force the further detection regions within some smaller ranges. In most traditional methods [4]–[8], the change map is directly obtained from the difference image, and the edges of the changed regions may be blurred consequently. Here, we make segmentation around the extracted keypoints in the two original multitemporal SAR images, where the edges of the detection regions are much clearer than those in the difference

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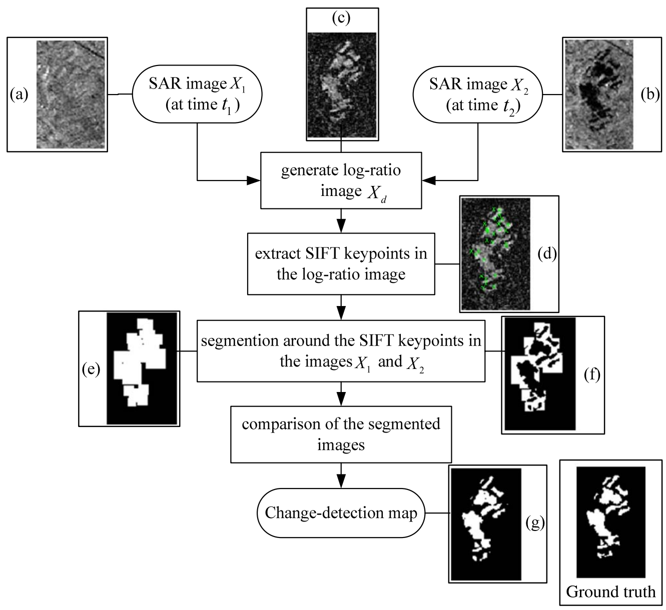


Fig. 1. General scheme of the proposed approach, where the example image clips are extracted from the image set of an area near the city of Bern, Switzerland, and will be illustrated in detail in Section III-A.

image, and further compare the two segmentations to generate the change-detection map.

The rest of this letter is organized as follows. Section II describes the proposed unsupervised change detection approach. Section III provides some experimental results of the proposed approach, and Section IV concludes this letter.

II. Proposed SAR Image Change Detection Technique

Let us consider two coregistered SAR images, $X_1 = \{x_1(i, j) | 1 \leq i \leq H, 1 \leq j \leq W\}$ and $X_2 = \{x_2(i, j) | 1 \leq i \leq H, 1 \leq j \leq W\}$, each with a size of $H \times W$, each acquired at the same geographical area but at different times, t_1 and t_2, respectively. We aim at generating a change-detection map representing changes that occurred between two images X_1 and X_2. As shown in Fig. 1, the proposed approach consists of four main steps: 1) generation of the log-ratio image; 2) extraction of the SIFT keypoints in the log-ratio image; 3) image segmentation around the extracted keypoints in the original input images; and 4) generation of the change-detection map by comparing the segmentation results. Example image clips are shown along each step for better illustration purpose. We mainly introduce steps 2) and 3) in the following.

A. Extraction of SIFT Keypoints in the Log-Ratio Image

To deal with speckle noise in the SAR image, the log-ratio operator, a well-known technique for producing a difference image [2], [4], [5], [8], is applied as the first step to generate the difference image. Let X_d be the log-ratio image, and it can be computed pixel by pixel from X_1 and X_2 by

$$X_d = \left| \log_{10} \left(\frac{X_2}{X_1} \right) \right|. \tag{1}$$

After obtaining the difference image, the next step is to analyze the difference image to discriminate changed regions from unchanged regions. As can be seen from (1), the values of pixels in changed regions will be larger than those in unchanged regions in X_d. That is to say, the changed regions to be detected are brighter than the background. In [12], the author defined the bloblike structure as the region which is brighter or darker than the background and stands out from its surroundings. Thus, the detection of changed regions from the difference image X_d can be seen as bloblike structure detection in the difference image where the main problem is to reduce the effect of speckle noise. As discussed in [11], SIFT can detect bloblike structures in the image and is robust to speckle noise. Therefore, we extract SIFT keypoints in the difference image to obtain the rough location and range of changed regions. Because of the introduction of the noise-robust SIFT keypoints, we can correctly reduce the changed regions of candidates to obtain high detection accuracy.

Lowe [13] proposed the SIFT to perform reliable image matching between different views of an object or scene. SIFT keypoints are first extracted in images, and each keypoint is described by a parameter set $\{x, y, \sigma, f\}$, where (x, y) represents the spatial coordinate of the keypoint, σ denotes the scale, and f, called the descriptor, is the 128-element feature vector calculated with the image sample points in a region around the keypoint location. In other words, SIFT first extracts bloblike structures from images and represents each bloblike structure with an appropriate feature. In our change detection task, we only exploit the location and scale information of SIFT keypoints in the log-ratio image to determine the location and range of the changed regions. We briefly explain the procedure of SIFT keypoint extraction in the following, and more details can be found in [13].

1) The scale-space extrema are detected as SIFT keypoint candidates in the difference scale space, $D(x, y, \sigma)$, which is generated by the difference of Gaussian (DoG) function convolved with image $I(x, y)$. The DoG function is the difference of two Gaussian functions with different scales σ and $k\sigma$, where k denotes a constant multiplicative factor

$$D(x, y, \sigma) = (G(x, y, k\sigma) - G(x, y, \sigma)) * I(x, y) \tag{2}$$

where

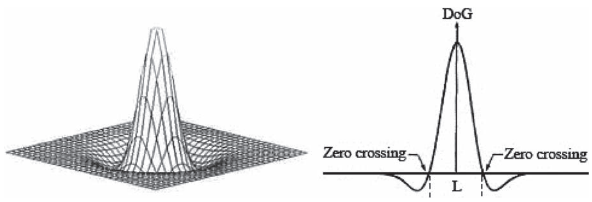
$$G(x, y, \sigma) = \frac{1}{2\pi\sigma^2} e^{-(x^2+y^2)/2\sigma^2}. \tag{3}$$

2) The unstable candidates, which have low contrast and which are localized along the edge, are rejected.

Fig. 1(d) shows the SIFT keypoints extracted in the log-ratio image. We can see that the SIFT keypoints are localized in the center of the bloblike structures and not affected by the speckle noise. Therefore, the following detection is forced within the small regions around the SIFT keypoints.

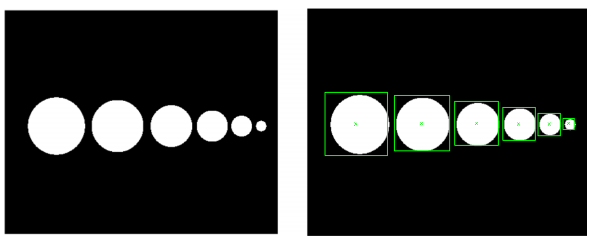
B. Segmentation Around the SIFT Keypoints in the Original Images

After extracting the SIFT keypoints, the next problem is how to obtain the changed regions based on these points. A straightforward idea is segmentation around the extracted SIFT keypoints to obtain the detailed changed regions. Given the



(a) (b)

Fig. 2. Three-dimensional plot and cross section of the negative of the DoG. (a) Three-dimensional plot of the negative of the DoG. (b) Cross section of (a).



(a) (b)

Fig. 3. Keypoint detection for a simple image. (a) Image to be detected. (b) SIFT keypoints detected in the image and ROCs around the SIFT keypoints.

location and scale information of SIFT keypoints, we can acquire regions of candidates (ROCs) which contain the changed regions.

Marr and Hildreth point out that, when $k = 1.6$, the DoG function can provide a close "engineering" approximation to the Laplacian of Gaussian (LoG) [14]. In the detailed experimental comparisons in [15], it is found that the maxima and minima of the LoG produce the most stable image features compared with some other functions widely used in the image processing. However, for its complexity, the LoG function is usually substituted by the DoG function. In this letter, we also use the DoG function with $k = 1.6$ to approximate the LoG. Fig. 2 shows the 3-D plot and cross section of the negative of the DoG function. The size of the DoG function, denoted as L, is defined as the distance between two zero crossings, and we can obtain the following equation:

$$L^2 = \frac{1}{8} \cdot \frac{(k\sigma)^2\sigma^2}{(k\sigma)^2 - \sigma^2} \ln\left(\frac{(k\sigma)^2}{\sigma^2}\right) = \frac{1}{8} \cdot \frac{k^2\sigma^2}{k^2 - 1} \ln(k^2). \quad (4)$$

It has been showed that the convolution response of a DoG function with an image attains an extremum when the size of a bloblike structure in the image is close to the size of DoG [15]. Therefore, the scale of the SIFT keypoint can be utilized to determine the size of the bloblike structure.

Substituting $k = 1.6$ into (4), we can infer that $L = 3.5\sigma$. Therefore, when the scale σ of a SIFT keypoint is known, we can draw a conclusion that the size of the bloblike structure around the SIFT keypoint is 3.5σ. Fig. 3 shows a simple example. Fig. 3(a) is the original image, and Fig. 3(b) shows the extracted SIFT keypoint (marked with x) and a green square with the size of $3.5\sigma \times 3.5\sigma$ centered at the SIFT keypoints. As can be seen from Fig. 3(b), the squares are exactly internal to the circle. Using the aforementioned relationship, we can obtain the ROCs around the SIFT keypoints. In order to obtain more accurate changed regions, we set the window size of ROCs as $4\sigma \times 4\sigma$ in the following experiments.

Owing to the fact that the edges of the changed regions in the difference image are blurred, conventional change detection methods, e.g., the generalized minimum-error and FCM methods, which directly segment the difference image, cannot preserve the edges of the changed regions well. Therefore, we deal with the original images, rather than the difference image, to acquire accurate changed regions. Details of the segmentation around the SIFT keypoints are listed in Algorithm 1 as follows.

Algorithm 1 Segmentation around the SIFT keypoints

Input: Original images X_1 and X_2. The number of SIFT keypoints: N. Location and scale set of SIFT keypoints: $\{x_i, y_i, \sigma_i\}_{i=1:N}$.

Output: segmentation result of image X_1: S_1, segmentation result of image X_2: S_2.

for $i = 1 \to N$ do

Extract the square region r_i^1 with its center at $\{x_i, y_i\}$ and of side $l_i^1 = 4\sigma_i$ in X_1.

Extract the square region r_i^2 with its center at $\{x_i, y_i\}$ and of side $l_i^2 = 4\sigma_i$ in X_2.

end for

Obtain ROCs of the image X_1: $R_1 = \{\cup r_i^1 | i \in \{1, \ldots, N\}\}$.

Obtain ROCs of the image X_2: $R_2 = \{\cup r_i^2 | i \in \{1, \ldots, N\}\}$.

Apply the Otsu [16] method to segment R_1 and R_2 and obtain the segmentation results S_1 and S_2.

Otsu is one of the most successful methods for image segmentation. It is an exhaustive algorithm of searching the global optimal threshold to minimize the within-class variance and has been widely used in the SAR image process [17], [18]. Usually, Otsu is directly used in the original SAR images. In the proposed method, we first determine the regions to be segmented, and then, Otsu is exploited, which greatly reduces the computation burden in the segmentation step.

Finally, by comparing the segmentation results S_1 and S_2, the change-detection map is obtained, as shown in Fig. 1(d).

III. Experiments

In this section, we demonstrate the performance of our proposed SAR image change detection method by presenting numerical results on four data sets.

A. *Experimental Image Set Description*

Four image sets are used for experiments in this letter, and each image set has been coregistered with single-pixel level accuracy to reduce the system error. The first image set is acquired by the European Remote Sensing 2 (ERS-2) satellite SAR sensor over an area near the city of Bern, Switzerland, in April and May 1999, respectively [see Fig. 4(a) and (b)]. Between the two dates, the River Aare flooded parts of the city. The image size is 301×301, and the available ground truth (reference image) is shown in Fig. 4(c). Another image set consists of two 256×256 SAR images [see Fig. 5(a) and (b)], acquired over an area in Indonesia by European Remote Sensing 1 (ERS-1) in February and March 1994, respectively. Ground changes are due to flood damage, and the available ground truth is shown in Fig. 5(c). The third set consists of two 423 × 322 SAR images [see Fig. 6(a) and (b)], acquired over a farmland in Italy on April 16 and 18, 1994, respectively. Ground changes are due to farmland evolution, and the available ground truth is shown in

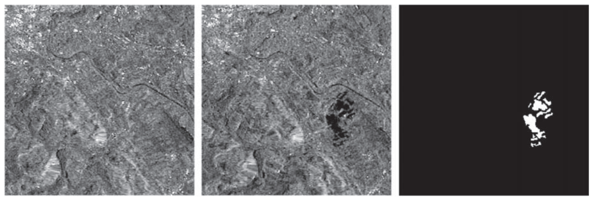

(a) (b) (c)

Fig. 4. SAR images relating to the city of Bern. (a) Before the flood. (b) After the flood. (c) Ground truth.

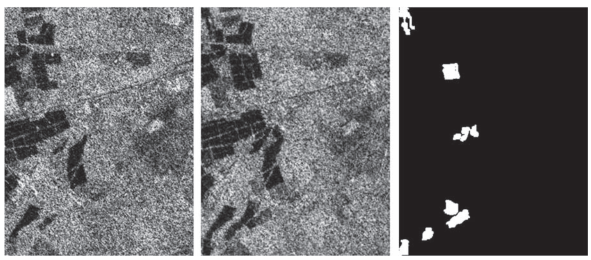

(a) (b) (c)

Fig. 5. SAR images relating to a farmland in Italy. (a) Before the flood. (b) After the flood. (c) Ground truth.

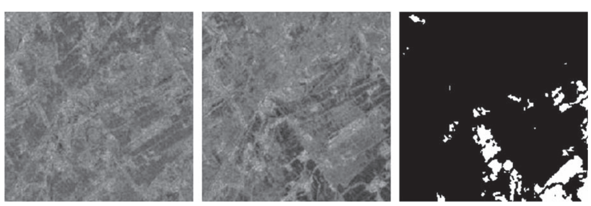

(a) (b) (c)

Fig. 6. SAR images relating to a farmland in Italy. (a) Before the flood. (b) After the flood. (c) Ground truth.

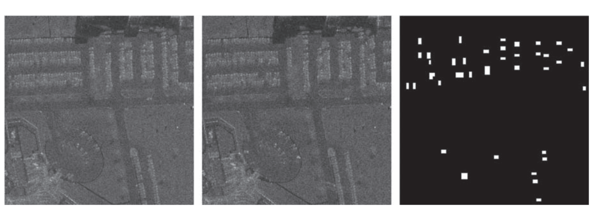

(a) (b) (c)

Fig. 7. SAR images relating to a parking lot. (a) Before some cars left. (b) After some cars left. (c) Ground truth.

Fig. 6(c). The last set is from an Air Force Research Laboratory (AFRL) airborne X-band sensor and consists of two 1000×1000 SAR images of a parking lot [see Fig. 7(a) and (b)]. In the interval of the two SAR images, some cars left the parking lot. The available ground truth is shown in Fig. 7(c).

B. Analysis on SIFT Keypoints

In order to illustrate the advantage of introducing the SIFT keypoints into SAR image change detection, the ROCs determined by the SIFT keypoints for the four data sets are shown in Fig. 8, and the area ratios of ROCs to the original images are given in Table I. From Table I, we can see that regions to be detected can be greatly reduced by extracting the SIFT keypoints. On the other hand, from Fig. 8, we can see that the ROCs almost contain the changed regions with little false alarms. Therefore, by extracting the noise-robust SIFT keypoints, we can correctly reduce the detection range of changed regions and thus ensure the final detection accuracy of the proposed method.

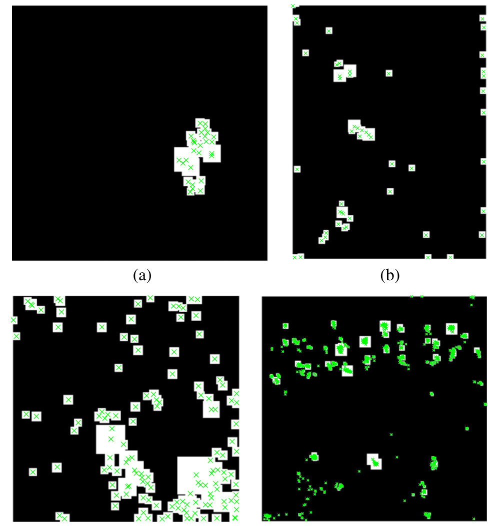

(a) (b)
(c) (d)

Fig. 8. ROCs determined by the SIFT keypoints for the four data sets (SIFT keypoints are marked with a green cross). (a) ROCs of Bern flooding. (b) ROCs of Indonesia flooding. (c) ROCs of Italy farmland. (d) ROCs of parking lot.

TABLE I
AREA RATIO OF ROCS TO THE ORIGINAL IMAGE

images	ratio
Bern flooding	3.13%
Indonesia flooding	4.55%
Italy farmland	22.02%
Parking lot	5.15%

C. Change Detection Results

We take the generalized minimum-error method [4] and the FCM method [8] as the reference methods in this letter to compare the detection performance with our proposed method. In the final results, black pixels on the change maps form the unchanged regions, whereas the bright pixels form the changed regions. False positives (FPs, unchanged pixels wrongly classified as changed), false negatives (FNs, changed pixels that are undetected), and the percentage of correct classification (PCC) are taken as the objective evaluation criteria to detection performance. FP corresponds to the false alarm, and FN corresponds to the missed alarm.

The change-detection maps of the four experiments are shown in Fig. 9. By comparing the change maps obtained by our proposed method, the generalized minimum-error method, and the FCM method with the reference images of the ground truth in Figs. 4(c), 5(c), 6(c), and 7(c), it is obvious that our proposed method yields more accurate changed regions and its false alarms mostly appear around the changed regions. Moreover, as reported in Table II, our proposed method results in the highest PCC. We can also see from Table II that the missed alarms (FN) of the generalized minimum-error method and the false alarms

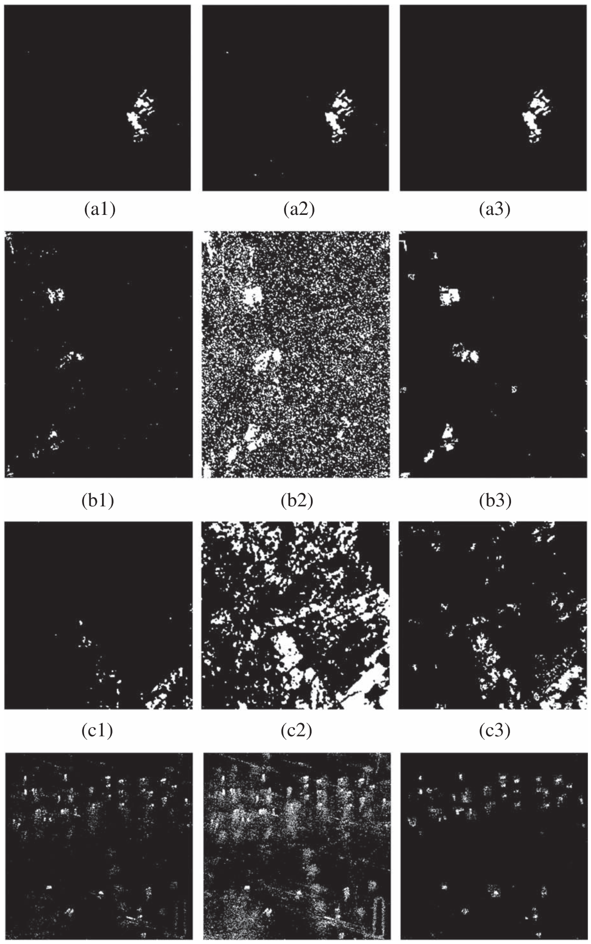


Fig. 9. SAR image change detection results. (a1, b1, c1, and d1) Change-detection maps resulted from the generalized minimum-error method. (a2, b2, c2, and d2) Change-detection maps resulted from the FCM method. (a3, b3, c3, and d3) Change-detection maps resulted from the proposed method.

TABLE II
CHANGE DETECTION RESULTS OF DIFFERENT METHODS

images	method	FP	FN	PCC
Bern flooding	Generalized Minimum-Error	36	343	99.58%
	FCM	258	72	99.64%
	proposed	113	187	**99.69%**
Indonesia flooding	Generalized Minimum-Error	192	3032	97.63%
	FCM	27289	890	79.30%
	proposed	2238	798	**97.77%**
Italy farmland	Generalized Minimum-Error	2	4755	92.74%
	FCM	7050	386	88.65%
	proposed	2360	1052	**94.29%**
Parking lot	Generalized Minimum-Error	11030	11445	97.76%
	FCM	84772	6803	90.60%
	proposed	6492	11121	**98.24%**

(FP) of the FCM method are high in all the four experiments, while our method has relatively low missed alarms and false alarms.

IV. Conclusion

In this letter, we propose a novel unsupervised distribution-free method for SAR image change detection, which is based on the SIFT keypoints and region information. The SIFT keypoints, which detect the bloblike structures and are robust to the speckle noise, can localize the range of the changed region, while image segmentation can reserve the region information of changed regions. Experimental results on real SAR data sets prove that our proposed method gives more precise results and exhibits stronger robustness against the speckle noise compared with the conventional methods.

References

[1] B. Hou, Q. Wei, Y. Zheng, and S. Wang, "Unsupervised change detection in SAR image based on Gauss-log ratio image fusion and compressed projection," *IEEE J. Sel. Topics Appl. Earth Observ. Sens.*, vol. 7, no. 8, pp. 3297–3317, Aug. 2014.

[2] H. Hu and Y. Ban, "Unsupervised change detection in multitemporal SAR images over large urban areas," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 8, pp. 3248–3261, Aug. 2014.

[3] L. Jia *et al.*, "Semisupervised SAR image change detection using a cluster-neighborhood kernel," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 8, pp. 1443–1447, Aug. 2014.

[4] G. Moser and S. B. Serpico, "Generalized minimum-error thresholding for unsupervised change detection from SAR amplitude imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 10, pp. 2972–2982, Oct. 2006.

[5] Y. Bazi, L. Bruzzone, and F. Melgani, "An unsupervised approach based on the generalized Gaussian model to automatic change detection in multitemporal SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 4, pp. 874–887, Apr. 2005.

[6] P. Lombardo and C. J. Oliver, "Maximum likelihood approach to the detection of changes between multitemporal SAR images," *Proc. Inst. Elect. Eng.—Radar, Sonar Navigat.*, vol. 148, no. 4, pp. 200–210, Aug. 2001.

[7] F. Chatеain, J. Y. Tourneret, and J. Inglada, "Change detection in multisensor SAR images using bivariate Gamma distribution," *IEEE Trans. Image Process.*, vol. 17, no. 3, pp. 249–258, Mar. 2008.

[8] J. J. Ma, M. G. Gong, and Z. Q. Zhou, "Change detection in synthetic aperture radar images based on image fusion and fuzzy clustering," *IEEE Trans. Image Process.*, vol. 21, no. 4, pp. 2141–2151, Apr. 2012.

[9] A. Ghosh, N. S. Mishra, and S. Ghosh, "Fuzzy clustering algorithms for unsupervised change detection in remote sensing images," *Inf. Sci.*, vol. 181, no. 4, pp. 699–715, Feb. 2011.

[10] T. Celik, "Unsupervised change detection in satellite images using principal component analysis and k-means clustering," *IEEE Geosci. Remote Sens. Lett.*, vol. 6, no. 4, pp. 772–776, Oct. 2009.

[11] K. Mikolajczyk and C. Schmid, "A performance evaluation of local descriptors," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 27, no. 10, pp. 1615–1630, Oct. 2005.

[12] T. Lindeberg, "Detecting salient blob-like image structures and their scales with a scale-space primal sketch: A method for focus-of-attention," *Int. J. Comput. Vis.*, vol. 11, no. 3, pp. 283–318, Dec. 1993.

[13] D. G. Lowe, "Distinctive image features from scale-invariant keypoints," *Int. J. Comput. Vis.*, vol. 60, no. 2, pp. 91–110, Nov. 2004.

[14] D. Marr and E. Hildreth, "Theory of edge detection," *Proc. R. Soc. London*, vol. B207, no. 1167, pp. 187–217, Feb. 1979.

[15] K. Mikolajczyk, "Detection of local features invariant to affine transformations," Ph.D. thesis, Dept. Comput. Vision Inst. Nat. Polytechnique de Greboble, Grenoble, France, 2002.

[16] N. Otsu, "A threshold selection method from gray-level histograms," *IEEE Trans. Syst. Man Cybern.*, vol. 9, no. 1, pp. 62–66, Jan. 1979.

[17] Z. Chen, C. Wang, X. Teng, L. Cao, and J. Li, "Oil spill detection based on a superpixel segmentation method for SAR image," in *Proc. IGARSS*, Quebec City, QC, Canada, Jul. 2014, pp. 1725–1728.

[18] Y. Yang, Y. Wang, and J. Qian, "Building identification from SAR image based on the modified marker-controlled watershed algorithm," in *Proc. IGARSS*, Milan, Italy, Jul. 2015, pp. 2481–2484.